Where does a longer think pay in rapid?

The app's long-think triggers on 74,000 Lichess 10-minute games

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Abstract

The app tells the player to stop and think at four kinds of moment: the first move out of the opening book (45 seconds), a winning position to convert, a forcing moment, and a pawn capture outside the endgame (30 seconds each). On 74,000 rated 10-minute Lichess games from 2026-09, built with the app's own code, players at his level made serious errors on 11.0% of ordinary moves played in under 10 seconds and on 23.1% of those played after 30 seconds or more. Compared only in positions a danger score rates alike, a long think was followed by more serious errors at ordinary moves (+4.7 points), at conversion moments (+8.8) and at forcing moments (+7.6), and by no fewer at the book exit (+1.6) or a pawn capture (+0.6). By the rule fixed before any effect was computed, no trigger shows that a long think pays, none of eight other moments does, and no think length is a bar. The pattern holds at every rating from 900 to 2099 and, at the book exit, within the very same position. Players think long where a position is harder than the board and the clocks show, and that outweighs whatever the think buys: this design cannot see past it. It does show two things the app can use. The triggers fire on 71% of the moves analysed and ask for about 541 of a game's 600 seconds, so they cannot be followed as written. And serious errors gather where the triggers do not point: a third of them come on the move right after the opponent's own serious error, while the book exit holds 1.6% of them.

1 The question

The app's method service (`triggerOf`, at commit `d8b771238c`) names the moments where a long think should pay. The first move out of the player's book should get 45 seconds. A position to convert (an evaluation of +3 or better), a forcing moment (in check, just after a capture, a piece hanging on either side, or a move that gives check) and a pawn capture outside the endgame should get 30. The app calls a move made in under 10 seconds rushed and one of 30 seconds or more a long think. For players at his level in 10-minute games, does a long think cut serious errors at these moments, and more than at an ordinary move? Do other moments pay more? Are 30 and 45 seconds the right bars?

2 The data

The Lichess database for 2026-09 (CC0) holds 6,185,560 games, 1,019,414 of them rated rapid; 195,409 of those were played at 10 minutes with no increment and carry Lichess's engine evaluation on every move beside the clock. The project's stream kept, in file order, the games that ended normally or on time, had at least 20 plies and two ratings within 300 points, by bins of the mean rating (Table 1).

mean rating	800–999	1000–1199	1200–1399	1400–1599	1600–1799	1800–1999	2000–2199
games	6,000	8,000	20,000	20,000	8,000	6,000	6,000

Table 1: Games kept, by the players' mean rating.

Each game was read as the routine's archive row and built with the app's own code through the lab's bridge: the metrics service's `primaryFacts` and `deriveGameFacts` with the routine's ports, and `triggerOf`. Every game was built once with each player as "me", so every move gets its moment from its own side; Black's build is White's with the evaluations and the result turned around, and on 800 games it gave every move the same facts as a real Black build. All 74,000 games built. Two choices are this study's. A Lichess player has no repertoire in the app, so the book exit is the first move the app's review reader does not call book (its book covers the first 12 plies). And a serious error is a mistake or a blunder as the reader classifies it from the evaluations; the app also counts misses, which need the routine's miss ledger.

The moves studied have a think time (each side's first move has none), are graded and not trivial (the app's `nonTrivial`: not book, not forced, a win probability before the move between 5 and 95%), and leave the mover at least 60 seconds on the clock before the move, so a long think was affordable: 3,738,138 of 4,721,351 timed moves. His chess.com rapid rating comes to about 1350 on Lichess by the app's default conversion, so the rule is applied to the moves of players rated 1250 to 1449: 980,946 moves in 39,902 player-games, of which 68% were quick and 6% long.

3 The method

Players think longer where a position is harder, and harder positions breed errors, so long and quick moves are compared only in positions a *danger score* rates alike. The score is the chance that a quick move in the position is a serious error, predicted from what the player could see before moving: both ratings and clocks, the opponent's last think, the ply and the moves since the book exit, the evaluation and win probability before the move, the phase, the material, the legal moves, the checks and captures available, the hanging pieces on each side, being in check, coming just after a capture or either side's serious error, and the colour. Nothing about the think or the move played. A gradient-boosted model (LightGBM, 400 trees, fixed settings) was trained on the quick moves of every rating and cross-fitted in two halves split by game, so each move's score comes from a model that never saw its game. It ranks quick moves well (AUC 0.799) and the others nearly as well (0.762).

For a set of moves T, its moves are split into five strata at the quintiles of the danger score over T's quick moves, and Δ_T is the long moves' serious-error rate minus the quick moves', averaged over the strata with weights equal to T's quick moves in each: how much a long think would change the error rate of the moves now played quickly. Intervals come from 2,000 bootstrap draws over games (Poisson weights, seed 20261014), the 95% percentile interval, and a p-value of twice the smaller share of draws on either side of zero; every comparison uses the same draws.

The rule, committed before any effect was computed: a long think *pays* at T when Δ_T's interval lies below zero; T *marks* where it pays when it also beats ordinary moves ($\Delta_T - \Delta_{\text{none}}$ below zero with a Holm-adjusted p under 0.05 over the four triggers). Each trigger is kept when both hold. Eight candidate moments, each defined before the move, would join the triggers under the same test over eight. A trigger's bar is the shortest think bin whose Δ reaches 80% of a 60-second think's, when

that one pays.

4 Results

4.1 The triggers

	per	moves		serious errors				Holm
moment	game	quick	long	quick	long	Δ	$\Delta - \Delta_{\text{none}}$	p
Book end	1.0	33,996	437	5.0	12.1	+1.6 (−0.1, +3.4)	−3.1 (−4.9, −1.2)	0.008
Conversion	5.3	134,722	13,192	11.9	23.4	+8.8 (+8.2, +9.5)	+4.1 (+3.3, +5.0)	0.004
Forcing	11.2	306,030	24,636	13.0	26.9	+7.6 (+7.1, +8.0)	+2.9 (+2.1, +3.6)	0.004
Structure	0.2	4,871	307	9.7	13.7	+0.6 (−2.3, +3.7)	−4.1 (−6.9, −1.0)	0.010
none	7.0	188,925	15,759	11.0	23.1	+4.7 (+4.1, +5.3)		

Table 2: The four triggers and ordinary moves (none), players rated 1250–1449. Per game: moves per player per game among those analysed. Quick: under 10 seconds; long: 30 seconds or more. Error rates are raw; Δ is long minus quick within danger strata, in percentage points, with its 95% interval.

At ordinary moves a long think is followed by +4.7 points more serious errors than a quick move in a position the score rates alike (Table 2). At conversion and forcing moments the excess is larger still (+8.8 and +7.6), and larger than at ordinary moves by +4.1 and +2.9 points. At the book exit and at a pawn capture, the intervals hold zero (−0.1 to +3.4 and −2.3 to +3.7), and the excess is smaller than at ordinary moves (by −3.1 and −4.1 points, Holm p 0.008 and 0.010). No trigger's interval lies below zero, so the verdict is the same for all four: *no sign it pays*. Read as the rule reports it, a long think's extra minute at a conversion moment goes with −0.128 serious errors saved (−0.137 to −0.118), at a forcing moment with −0.106: a cost, not a saving.

4.2 Eight other moments

	per	moves				Holm	
before the move	game	quick	long	Δ	$\Delta - \Delta_{\text{none}}$	p	joins
in check	1.5	50,679	959	+3.7 (+1.7, +5.7)	−1.0 (−3.1, +1.0)	0.648	no
after a capture	6.3	194,911	9,034	+8.2 (+7.5, +8.9)	+3.5 (+2.6, +4.4)	0.008	no
his piece hanging	9.4	241,606	25,008	+8.7 (+8.2, +9.2)	+4.0 (+3.3, +4.7)	0.008	no
opponent's piece hanging	6.3	186,669	11,791	+10.6 (+9.9, +11.3)	+5.9 (+5.0, +6.8)	0.008	no
after the opponent's error	3.4	85,675	8,489	+16.3 (+15.1, +17.5)	+11.6 (+10.4, +12.8)	0.008	no
defending, −3 or worse	4.5	116,842	11,270	+2.3 (+1.9, +2.7)	−2.4 (−3.0, −1.8)	0.008	no
book end and his next four	4.8	159,050	4,040	+4.6 (+3.7, +5.5)	−0.1 (−1.0, +0.8)	0.901	no
high danger, top fifth	5.7	133,709	18,266	+12.9 (+12.2, +13.7)	+8.2 (+7.4, +9.1)	0.008	no

Table 3: The candidates, each defined by what the player sees before moving. Joins: pays, and beats ordinary moves with a Holm-adjusted p under 0.05 over the eight.

None pays (Table 3): at every candidate a long think is followed by more serious errors than a quick one, most of all right after the opponent's serious error (+16.3) and in the top fifth of the danger

score (+12.9). The first four candidates are the parts of the forcing trigger; none of them carries it.

4.3 The bars

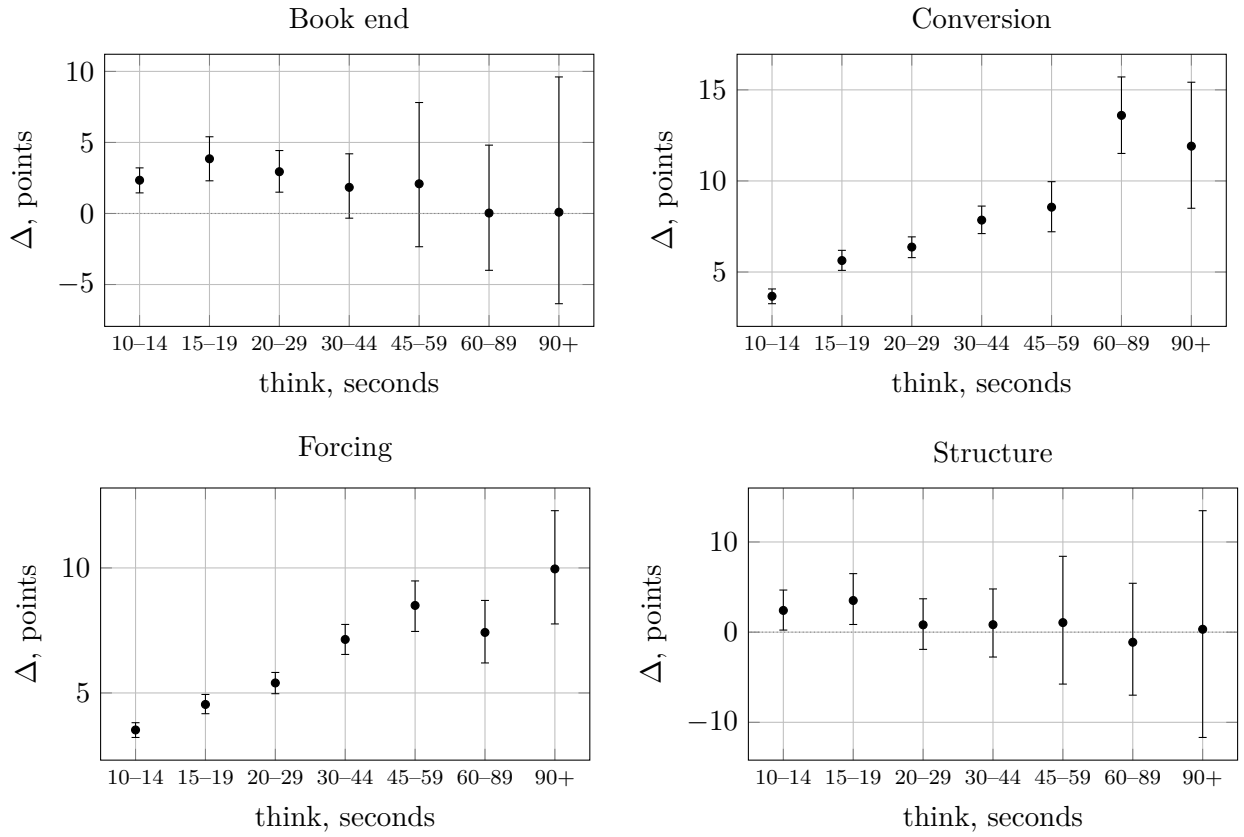


Figure 1: Δ for each length of think against the quick moves, by trigger, in percentage points with 95% intervals. Below zero would mean fewer serious errors after the longer think.

No think length pays at any trigger, so no trigger has a bar (Figure 1). At conversion and forcing moments the excess grows with the length of the think: every bin from 30 seconds up lies above the 10–14-second bin. The longer the think, the harder the position it marks, in ways the danger score does not see.

4.4 Every rating

The answer does not move with the level (Table 4): at every rating from 900 to 2099, ordinary, conversion and forcing moves are followed by more serious errors after a long think. So the uncertainty in his Lichess level matters little here.

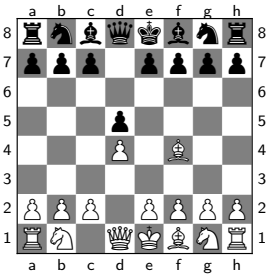
4.5 The same position

A book-exit move is often made in a position many games reach, so long and quick moves can be compared in the very same position, with no danger score at all: 292 positions were reached with

mover's rating	none	book end	conversion	forcing	structure
900–1099	+5.3 (+4.2, +6.4)	+8.8 (+4.3, +14.2)	+10.0 (+8.9, +11.1)	+7.8 (+6.9, +8.7)	+1.1 (−5.6, +7.6)
1100–1299	+4.9 (+4.2, +5.6)	+2.8 (+0.7, +5.0)	+8.7 (+7.9, +9.4)	+7.6 (+7.0, +8.2)	+3.9 (−0.4, +8.3)
1300–1499	+4.6 (+4.0, +5.1)	+2.2 (+0.3, +4.2)	+8.9 (+8.2, +9.6)	+7.7 (+7.2, +8.1)	−0.1 (−3.0, +3.1)
1500–1699	+4.5 (+3.9, +5.1)	+1.2 (−0.7, +3.3)	+8.8 (+8.1, +9.6)	+8.2 (+7.6, +8.7)	+3.8 (+0.3, +7.5)
1700–1899	+5.0 (+4.3, +5.7)	+1.3 (−0.8, +3.8)	+8.9 (+7.9, +10.0)	+7.1 (+6.4, +7.8)	−3.0 (−6.0, +0.5)
1900–2099	+4.0 (+3.4, +4.6)	+0.8 (−1.0, +2.9)	+8.8 (+7.7, +9.9)	+7.4 (+6.8, +8.1)	+5.8 (+1.4, +10.7)

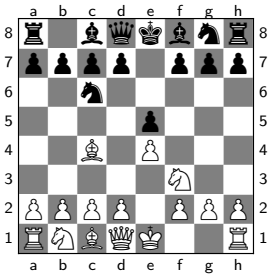
Table 4: Δ by the mover's rating, in percentage points with 95% intervals. Described, not tested.

both (9,974 quick moves, 386 long). Weighted by their quick moves, the long thinks were followed by +1.9 points more serious errors (−1.0 to +5.0): a cut of more than 1.0 points is ruled out. The positions most games reached are quiet (Figure 2): in them long thinks are rare, and serious errors rarer than at his level's average move.



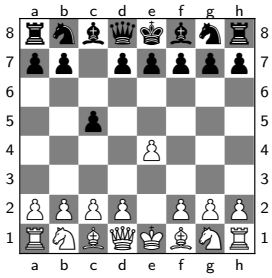
Black to move: 655 quick moves, 0.0% serious; 6 long, 0.0%.

`rnbqkbnr/ppp1pppp/8/3p4/3P1B2/8/PPP1PPPP/RN1QKBNR b KQkq -`



Black to move: 477 quick moves, 6.9% serious; 4 long, 0.0%.

`r1bqkbnr/pppp1ppp/2n5/4p3/2B1P3/5N2/PPPP1PPP/RNBQK2R b KQkq -`



White to move: 324 quick moves, 0.0% serious; 1 long, 0.0%.

`rnbqkbnr/pp1ppppp/8/2p5/4P3/8/PPPP1PPP/RNBQKBNR w KQkq -`

Figure 2: The three book-exit positions the most games at his level reached, with the moves made in them.

4.6 Sensitivity

With ten strata instead of five, and with quick moves limited to those over 5 seconds (the app's own clock buckets treat faster moves as decided before the turn), no trigger pays and each comparison with ordinary moves keeps its side (Table 5).

	ten strata		quick from 6 seconds	
moment	Δ	$\Delta - \Delta_{\text{none}}$	Δ	$\Delta - \Delta_{\text{none}}$
Book end	+1.2 (−0.4, +3.0)	−3.6 (−5.3, −1.8)	+0.5 (−1.7, +2.8)	−3.5 (−5.8, −1.2)
Conversion	+8.8 (+8.1, +9.4)	+3.9 (+3.1, +4.8)	+7.2 (+6.5, +8.0)	+3.2 (+2.3, +4.2)
Forcing	+7.4 (+6.9, +7.9)	+2.6 (+1.9, +3.3)	+6.4 (+5.9, +7.0)	+2.4 (+1.6, +3.2)
Structure	+0.8 (−2.2, +3.9)	−4.0 (−7.0, −0.9)	−1.2 (−4.8, +2.7)	−5.2 (−8.9, −1.3)

Table 5: The triggers' Δ and $\Delta - \Delta_{\text{none}}$ under two other choices.

5 Where the errors are

This section was added after the rule was committed, before the month's results were computed, and tests nothing. A trigger can save at most the errors made at its moment, so it is worth knowing where they are. At his level a player makes about 3.6 serious errors a game among the moves analysed (14.5% of them). Table 6 sets each moment's share of the moves against its share of those errors.

moment	moves	share of moves	share of serious errors	ratio
book end	38,411	3.9	1.6	0.40
conversion	209,836	21.4	21.2	0.99
forcing	445,443	45.4	49.9	1.10
structure	6,929	0.7	0.5	0.76
none	280,327	28.6	26.8	0.94
in check	61,508	6.3	4.4	0.70
after a capture	253,356	25.8	22.5	0.87
his piece hanging	375,658	38.3	44.1	1.15
opponent's piece hanging	252,969	25.8	26.6	1.03
after the opponent's error	133,873	13.6	33.1	2.43
defending, −3 or worse	178,114	18.2	6.0	0.33
book end and his next four	193,286	19.7	12.4	0.63
high danger, top fifth	227,445	23.2	54.8	2.37

Table 6: Each moment's share of the moves analysed at his level and of their serious errors, at every length of think. Ratio: the second over the first. The candidates overlap.

The right moment to stop is where the errors are. The move right after the opponent's serious error is 14% of the moves and 33% of the serious errors, about 1.2 a game. By the evaluations, failing to take what an error offers costs as much as making one, so these include every punishment missed. The top fifth of the danger score holds 55%. The book exit, where the app asks for its longest think, is 3.9% of the moves and 1.6% of the errors, about 0.06 a game: even a think that removed every one of them would save little. Meanwhile the triggers fire about 18 times in a player's game, on 71% of the 25 or so moves analysed (the forcing trigger alone on 45%), and at the app's bars they ask for about 541 seconds, 90% of a 10-minute game.

6 His games

His games meet the moments about as often as his level does (Table 7), except the rare pawn capture outside the endgame, which comes up more often in his. He thinks long on 8.3% of his conversion moves, against 6.3% at his level; at the other moments his share is close to theirs. After a long think his own serious errors run above those after a quick move at every moment with at least 50 long thinks to read, as at his level.

	per game		share of long thinks		his serious errors	
moment	his	his level	his	his level	quick (n)	long (n)
book end	0.9	1.0	1.4	1.1	1.5 (525)	0.0 (8)
conversion	4.8	5.3	8.3	6.3	8.6 (1,735)	18.2 (253)
forcing	12.2	11.2	6.1	5.5	9.7 (5,040)	23.7 (473)
structure	0.2	0.2	3.8	4.4	10.3 (107)	33.3 (6)
none	7.6	7.0	5.7	5.6	6.9 (3,163)	16.7 (270)

Table 7: His 632 10-minute games in the app's archive, read the same way and only described. His error rates come from the routine's engine rather than Lichess's, so they are not set beside his level's.

7 What it means for the app

1. **The triggers' think times are advice, not a measurement.** Nothing here shows a long think cutting serious errors at any moment, for any rating. That does not show the advice is wrong: it shows that where players choose to think long, the position is harder than anything the board and the clocks reveal, by more than the think can make up. The app should not tell him a think time at a moment saves errors as if it had been measured.
2. **Fewer triggers.** As written they fire on most of the moves he plays and ask for nine tenths of his clock. A trigger list a player can follow in 10 minutes has to be short, and the forcing trigger, which fires on almost half the moves, is the place to cut.
3. **Point them where errors are.** The move after the opponent's serious error holds a third of the serious errors at his level on a seventh of the moves; it is also the ledger's own "after a gift" leak. The book exit holds almost none: its 45 seconds are the cheapest to give back.
4. **Measure the effect with players, not positions.** Whether a long think pays at a moment can be read by comparing players who habitually think longer there with others of their rating, in their other games, against the same comparison at ordinary moves; or, on his own games, by having the coach flag a moment in some games and not in others.

8 Limits

The study is observational. The danger score adjusts for what the board and the clocks show, not for what the player saw; the comparison with ordinary moves is less exposed when the bias is alike, but it is not alike at every moment. Two parts of the triggers depend on the move played (giving check, a pawn capture), which a long think may change. The clock a long think takes from later moves is not counted. The book exit is the reader's 12-ply book's, not a repertoire's, and a serious error leaves out misses. Lichess evaluations come from its own server analysis; games someone asked to have analysed are not a random sample; and his Lichess level is the app's conversion of his chess.com rating, not a measurement, though the answer is the same at every level.